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Loehl

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(54) **ROUND FILTER ELEMENT**

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B01D 2275/206 (2013.01);

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See application file for complete search history.

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(57) **ABSTRACT**

A round filter element has an annular circumferentially
extending pleated filter medium body to be flowed through
by a fluid to be purified radially relative to a longitudinal
axis of the filter medium body. A first end disk is arranged
at a first end face of the filter medium body. A second end
disk is arranged at a second end face of the filter medium
body. The second end face is axially opposite the first end
face. An outer or an inner contour of the first end disk is
identically or at least substantially identically embodied to
an outer or an inner contour of the filter medium body. An
outer or an inner contour of the second end disk is identically
or at least substantially identically embodied to the outer or
the inner contour of the filter medium body. One of the first
and second end disks is open.

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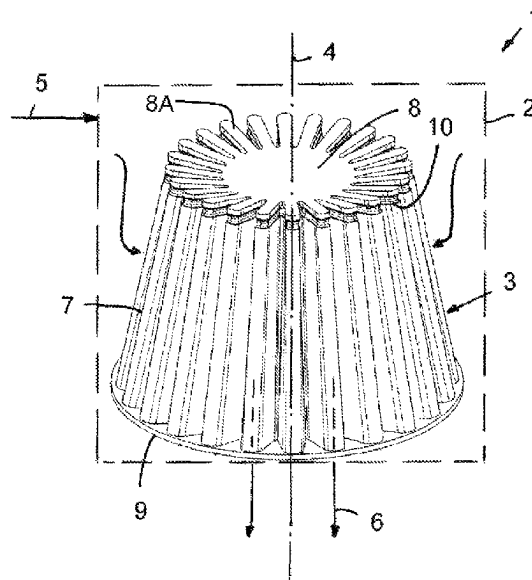
B29C 45/14 (2006.01)

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16 Claims, 3 Drawing Sheets



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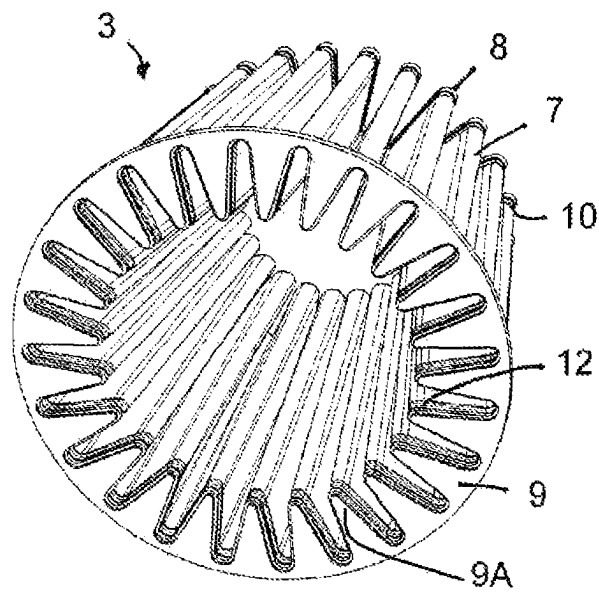
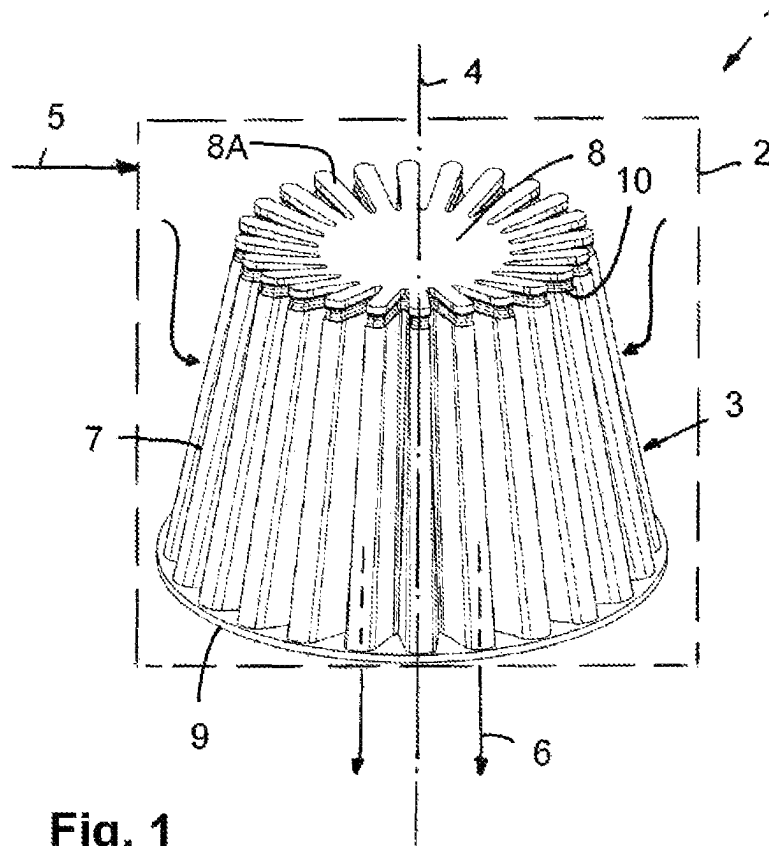
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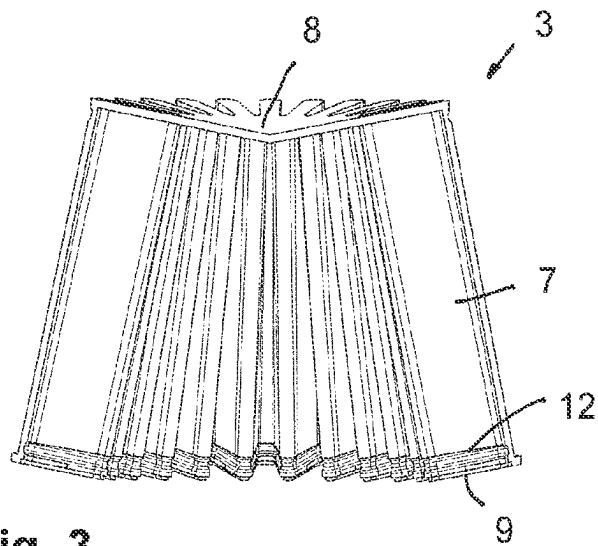


Fig. 3

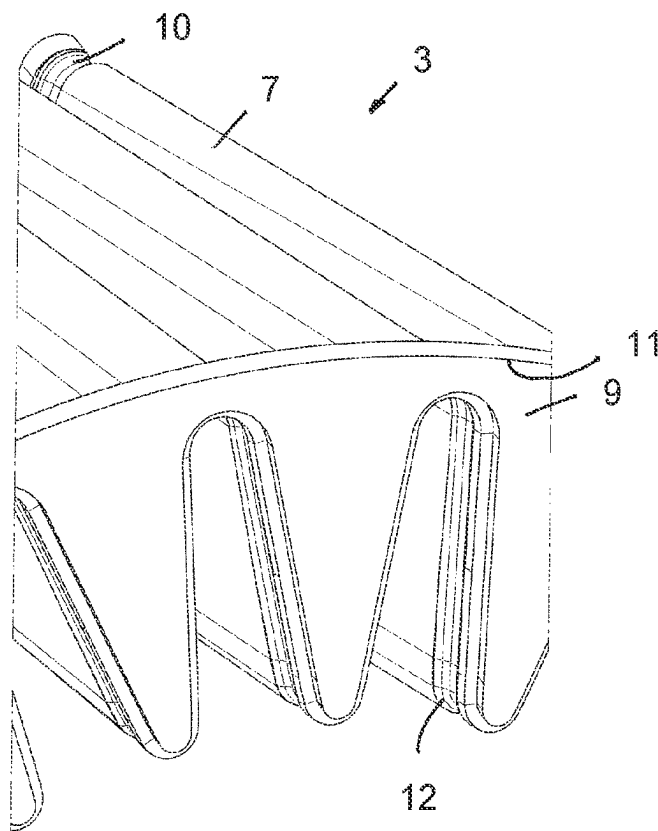


Fig. 4

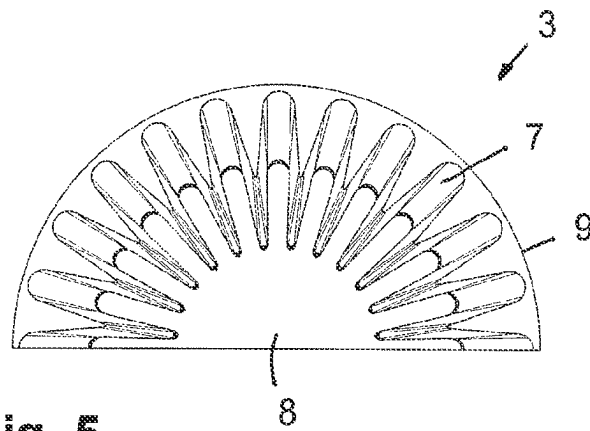


Fig. 5

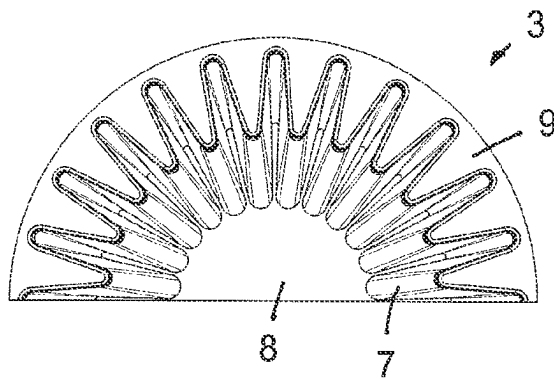


Fig. 6

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ROUND FILTER ELEMENT**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a continuation application of U.S. patent application Ser. No. 17/172,198 filed 10 Feb. 2021, which is a continuation of international application No. PCT/EP2019/070855 having an international filing date of 2 Aug. 2019 and designating the United States, the international application claiming a priority date of 14 Aug. 2018 based on prior filed German patent application No. 10 2018 119 683.0, the entire contents of the aforesaid U.S. patent application, the aforesaid international application and the aforesaid German patent application being incorporated herein by reference to the fullest extent permitted by the law.

BACKGROUND OF THE INVENTION

The invention relates to a round filter element, in particular for gas filtration, for example, for an air filter, with an annular circumferentially extending pleated-type filter medium body which can be flowed through by a fluid to be purified in radial direction in relation to the longitudinal axis, comprising an end disk at an end face of the filter medium body.

DE 11 2013 005 697 T5 discloses a round filter element for an air filter, wherein the round filter element comprises an approximately hollow cylindrical filter medium body with filter pleats that is flowed through by the air to be purified in radial direction. For supporting the filter medium body, a support frame with longitudinal and circumferential webs is placed on the filter medium body wherein at the end face of the support frame a closed end disk is arranged which, in the mounted state, contacts an end face of the filter medium body. The closed end disk at the support frame comprises a star-shaped cross section shape that corresponds to the cross-section shape of the filter medium body at its end face in such a way that the outer contour of the end disk coincides with the outer contour of the filter medium body at its end face. At the oppositely positioned end face that is of an open configuration, a sealing ring is provided at the support frame.

It is the object of the invention to configure a round filter element with an annular circumferentially extending pleated-type filter medium body with high efficiency with simple constructive measures.

SUMMARY OF THE INVENTION

This object is solved according to the invention in that an end disk is arranged, respectively, at axially oppositely positioned end faces of the filter medium body, wherein one end disk is of an open configuration, and in that the outer or the inner contour of the end disk is identically or at least substantially identically embodied to the outer or the inner contour of the pleated-type filter medium body.

The dependent claims provide expedient further developments.

The round filter element according to the invention is preferably used for gas filtration, for example, for an air filter for filtration of the combustion air which is supplied to an internal combustion engine. However, an application of the round filter element for liquid filtration is also conceivable.

The round filter element comprises an annular circumferentially extending filter medium body which is embodied as a hollow body and provided with filter pleats, wherein the

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flow of the fluid to be purified through the filter medium body—in relation to the longitudinal axis of the round filter element—is realized in radial direction. Advantageously, the filter medium body is flowed through by the fluid in radial direction from the exterior to the interior so that the outer side of the filter medium body forms the raw side and the inner side the clean side; the outflow of the purified fluid from the interior enclosed by the filter medium body is realized in axial direction. In the context of the invention, a flow through the filter medium body in radial direction from the interior to the exterior is also conceivable however.

For the annular circumferentially extending filter medium body, circular cross section shapes as well as cross section shapes that deviate from the circular shape are conceivable, for example, oval cross section shapes.

The filter pleats of the filter medium are advantageously oriented such that the pleat edges of the filter pleats extend in axial direction or are positioned in relation to the longitudinal axis at a preferably small angle of maximally 30°.

Embodiments are thus also conceivable in which the filter medium body has a spiral-shaped or twisted configuration so that the filter edges of the pleated filter along the outer side of the filter medium body extend in a spiral shape. This is achieved, for example, in that the axially oppositely positioned end faces of the filter medium body are twisted relative to each other about the longitudinal axis of the filter medium body.

The filter medium body is provided at its two axially oppositely positioned end faces with an end disk, respectively. The end disks are in particular fixedly connected to the filter medium body. A first end disk is preferably of a closed configuration so that via this end disk no axial flow of the fluid is possible between the interior enclosed by the filter medium body and the surroundings outside of the filter medium body. The second oppositely positioned end disk is of an open configuration and enables an axial flow of the fluid that, for a radial flow through the filter medium body from the exterior to the interior, is axially discharged in axial direction through the open end disk.

As needed, both end disks can be of an open configuration wherein in this case the opening in one end disk is closed off by a housing-associated counter contour in the installed state.

The filter pleats extends across the entire axial length of the filter medium body so that the filter medium body in the region of both axial end faces comprises a pleated-type inner and outer contour. According to the invention, the outer or inner contour of the end disks is identical to or at least substantially the same as the outer or the inner contour of the pleated-type filter medium body in the respective end face section.

This configuration has the advantage that, due to the adaptation of the end disk contour to the end edge contour of the filter medium body, flow resistances of the fluid flowing through the round filter element can be reduced. Also, the use of end disk material can be reduced.

Advantageously, the open end disk follows the inner contour of the filter medium body so that the inner contour of the open end disk coincides completely or at least approximately with the inner contour of the filter medium body. The open end disk provides thus no flow resistance for the fluid when flowing through in axial direction.

According to a further advantageous embodiment, the closed end disk follows the outer contour of the filter medium body. This embodiment has the advantage that the outer contour of the closed end disk corresponds to the outer contour of the filter medium body and does not project past

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the outer contour of the filter medium body so that a space-saving arrangement is provided.

According to a further advantageous embodiment, the open end disk whose inner contour corresponds to the inner contour of the filter medium body comprises an annular circumferentially extending outer contour. This annular circumferentially extending outer contour comprises a deviating contour relative to the outer contour of the filter medium body. Accordingly, the open end disk is matched only at its inner contour to the filter medium body. The annular circumferentially extending outer contour at the open end disk makes it possible, for example, to arrange a circumferentially extending sealing ring at the open end disk which is positioned in particular at the axially externally positioned side face of the open end disk adjacent to the annular circumferentially extending outer contour. As needed, the sealing ring can also be arranged at the radially externally positioned outer side of the annular circumferentially extending outer contour.

According to yet a further advantageous embodiment, the filter medium body comprises a conical shape so that the axially oppositely positioned end faces of the filter medium body comprise a differently sized diameter. The cone angle—the angle deviation in relation to the longitudinal axis of the filter medium—lies in particular in an angle range of maximally 30°, for example, maximally 25°, maximally 20°, maximally 15°, maximally 10°, or maximally 5°. The cone-shaped embodiment of the filter medium body enables in particular an improved inflow. For example, upon flow through the filter medium in radial direction from the exterior to the interior the fluid to be purified is supplied axially to the outer side of the filter medium body wherein the cone shape promotes a radial deflection to the interior. Moreover, it is possible to use the round filter element with the cone-shaped filter medium body, for example, in gas-conducting lines without a separate filter housing. Also, additional installation spaces in receiving filter housings can be made available due to the cone shape.

Also, a hollow cylindrical filter medium body is conceivable as an alternative to a cone-shaped filter medium body.

Within a conduit, for example, an air-guiding one, the filter element can optionally be used as an HC trap for binding entrained hydrocarbons.

The invention relates furthermore to a filter device with an afore described round filter element wherein the filter device comprises a filter housing for receiving the round filter element.

The invention relates also to a method for producing an afore described round filter element, the method including providing a filter medium; forming a plurality of pleats into the filter medium; forming the pleated filter medium into an annular pleated filter medium body, the annular pleated filter medium body is circumferentially closed and forms a hollow body having a hollow interior, such that the annular pleated filter medium body surrounds a longitudinal axis extending through a hollow interior of the annular pleated filter medium body from a first axial end face to an opposite second axial end face of the annular pleated filter medium body. Individual pleats of the plurality of pleats of the annular pleated filter medium body extend axially from the first axial end face to the opposite second axial end face of the annular pleated filter medium body.

As a note-axial, as used herein, is a direction of the longitudinal axis. radial, as used herein, is a direction transverse to the longitudinal axis.

The method preferably further includes axially compressing the filter medium of the annular pleated filter medium

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body, to form a compressed section arranged adjacent to but spaced axially inwardly away from a respective one of the axial end faces by a distance of maximally 5 mm, forming an uncompressed free region of the filter medium between the compressed section and the respective axial end face. Forming and fixing a first end disk onto an axial end face of the annular pleated filter, wherein the forming and fixing of the first end disk is carried out by: providing a flowable plastic material; applying the flowable plastic material in a flowable state onto the uncompressed free region of the annular filter medium body such that the compressed section forms a barrier separating the flowable plastic material from an axially inward uncompressed region of the annular pleated filter medium body; and curing or hardening the flowable plastic material on the axial end face of the annular pleated filter medium body, forming the first end disk.

The method may further include forming and fixing a second end disk onto an opposite axial end face of the annular pleated filter, the second end disk, may be formed as described for the first end disk.

Preferably the outer contour or an inner contour of the first end disk is identically or at least substantially identically embodied to an outer contour or an inner contour of the annular pleated filter medium body. Preferably the outer contour or inner contour of the second end disk is identically or at least substantially identically embodied to the outer contour or the inner contour of the annular pleated filter medium body.

The method preferably may further include forming the pleated filter medium body such that the first axial end face has a smaller outer diameter than the opposite second axial end face such that the annular pleated filter medium body has an outer diameter that changes conically from the first axial end face to the opposite second axial end face at a cone angle of maximally 30 degrees.

In the applying the flowable plastic material, the flowable plastic material may be applied by either: immersing the uncompressed free region of the annular filter medium body into the flowable plastic material, or injection molding the flowable plastic material onto the uncompressed free region of the annular filter medium body.

Preferably, one of the first and second end disks is an open end disk and a different one of the first and second end disks is a closed end disk.

In the forming an end disk, preferably an outer contour of the closed end disk follows an adjacent outer contour of the annular filter medium body. Preferably an inner contour of the open end disk follows an adjacent inner contour of the filter medium body.

In the method, axially compressing the filter medium is preferably includes: providing a lower casting mold part and an upper casting mold part; and placing the annular pleated filter medium body onto the lower casting mold part. Preferably the axially compressing the filter medium is practiced by: moving the upper casting mold part from above onto the annular pleated filter medium body and moving the upper casting mold part towards the lower casting mold part, wherein one of the casting mold parts is positioned inside the annular filter medium body and a different one of the casting mold parts is positioned outside of the filter medium body. The practice further includes interacting the upper casting mold part with the lower casting mold part such that the compressed section is formed in the annular filter medium body adjacent to but spaced axially inwardly away from a respective one of the axial end faces by a distance of maximally 5 mm.

The invention relates also to a method for producing an afore described round filter element by means of the two or more casting mold parts that are inserted from axially opposite directions in and about the annular circumferentially extending pleated-type filter medium body or placed on the latter. The two casting parts are located at the inner side and the outer side of the filter medium body and interact in this context in such a way with each other that the material of the filter medium body is compressed adjacent to an end face but at a distance to this end face. After introducing the casting mold parts and compression of the material of the filter medium body, an end disk is subsequently formed in that the end disk material in a flowable state is applied to that end face whose material is compressed by the two interacting casting mold parts. The end disk material can be injection molded around or the end edge of the filter medium body is immersed in a mold with liquid or at least flowable end disk material.

This procedure has the advantage that only a single process step is required in order to fasten the end disk at the filter medium body. By compression of the end face filter medium body material, it is prevented that the end disk material in the liquid or at least flowable state spreads into the compressed section of the filter medium body. It is thus ensured that the end disk material penetrates only in a defined region into the material of the filter medium body and produces a fixed connection with the filter medium body.

The end disk material is in particular a plastic material. The filter medium body is comprised, for example, of nonwoven material.

The method for producing the round filter element is suitable in particular for cone-shaped filter medium bodies in which an axial interaction of the casting mold parts with axial approach inside or outside of the filter medium body can be easily realized.

Alternatively, the method for producing a round filter element can be used also for hollow cylindrical filter media bodies.

The compression of the filter medium body material by means of the two casting mold parts is realized adjacent to the end face, but at the same time at a minimal distance to the end face, in order to enable immersion of the end face into the liquid or flowable end disk material during manufacture and to permit spreading of the end disk material into the uncompressed region. The free region that is neighboring the end face and is not compressed has an axial extension of, for example, maximally 5 mm or maximally 3 mm.

The end disk material can be introduced in liquid or at least flowable form into a hollow space which is provided in a casting mold part. During manufacture, first the filter medium body is placed onto a casting mold part positioned at the bottom, subsequently the second casting mold part is positioned axially on the filter medium body as well as the lower casting mold part, whereupon the desired compression of the filter medium body material adjacent, but at a distance, to the bottom end face of the filter medium body is realized. Subsequently, the liquid or flowable end disk material can be introduced into the cutout in the lower casting mold part so that the end disk material comes into contact with the uncompressed part of the filter medium body at the end face and can cure.

Advantageously, end disks are cast or injection molded on the filter medium body at both oppositely positioned end faces, wherein at a minimal distance to the respective end face the filter medium body material is compressed by means of the two interacting casting mold parts.

BRIEF DESCRIPTION OF THE DRAWINGS

Further advantages and expedient embodiments can be taken from the additional claims, the figure description and the drawings.

FIG. 1 shows a perspective view of a filter device with a round filter element that comprises a conical pleated-type filter medium body with end disks applied to the end faces.

FIG. 2 shows a perspective view of the open end face of the round filter element.

FIG. 3 shows the round filter element in half section view.

FIG. 4 shows an enlarged detail view of the open end disk at the filter medium body.

FIG. 5 shows the round filter element in plain view with closed end disk.

FIG. 6 shows the round filter element in a view from below with the open end disk.

In the Figures, same components are provided with same reference characters.

DESCRIPTION OF PREFERRED EMBODIMENTS

In FIG. 1, a filter device 1 is illustrated that can be used for gas filtration, for example, for an air filter for filtration of the combustion air of an internal combustion engine. The filter device 1 comprises a filter housing 2 and a round filter element 3 arranged in the filter housing 2 that, in relation to its longitudinal axis 4, is flowed through by the fluid to be purified in radial direction from the exterior to the interior. The fluid is introduced in the unpurified state via the fluid inlet 5 into the filter housing 2 and discharged in the purified state from the filter housing 2 through a fluid outlet 6. The outer side of the round filter element 3 forms the raw side, the inner side of the round filter element 3 forms the clean side that surrounds a preferably cylindrical interior inside the round filter element 3. From the interior in the round filter element 3, the purified fluid can be discharged axially through the fluid outlet 6 from the round filter element 3 and the filter device 1.

As can be seen in FIG. 1 in connection with the additional Figures, the round filter element 3 comprises a filter medium body 7 and end disk 8 and 9 at the oppositely positioned end faces of the filter medium body 7. The filter medium body 7 is embodied as an annular circumferentially extending body in cone shape and comprises a plurality of filter pleats whose filter edges form pleat tips (7A, 7B) extending at the cone angle of the cone shape in relation to the longitudinal axis 4 of the filter medium body 7. The cone angle amounts to 15° to 20° in the embodiment. The filter pleats of the filter medium body 7 are formed across the axial length of the filter medium body in a steady shape so that the pleat height—viewed in the radial direction—along the axial height of the filter medium body 7 remains the same. The filter pleats increase in a significant manner the filter surface of the filter medium body 7. The filter medium body 7 is comprised, for example, of a nonwoven material.

The end disk 8 at the narrower end face of the filter medium body 7 is of a closed configuration while the end disk 9 at the wider end face of the filter medium body 7 is of an open configuration. The end disks 8 and 9 are comprised of a plastic material that, for manufacturing, is introduced in liquid or at least flowable state into a casting mold part into which the respective end face region of the filter medium body 7 is immersed. In this context, two casting mold parts interact of which the lower casting mold part is provided with a cutout for receiving the end disk

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material. During manufacture, first the filter medium body continuous manner so that the fold is placed onto the lower casting mold part without end disk material, subsequently a second casting mold part is placed from above onto the filter medium body and the lower casting mold part, wherein one casting mold part is positioned inside the filter medium body and the second casting mold part outside of the filter medium body. The casting mold parts interact such with each other in such a way that the material of the filter medium body adjacent to the end face, but with a minimal spacing to the end face, is compressed.

After introducing the liquid or at least flowable end disk material into the correlated cutout in the lower casting mold part, the uncompressed section of the filter medium body adjoining immediately the end face is immersed in the end disk material that can penetrate into the filter medium body material so that, after curing of the end disk material, a fixed connection between the end disk and the filter medium body is provided. At the same time, the compressed material of the filter medium body prevents undesirable spreading of end disk material into the filter medium body. This procedure can be carried out at both end faces of the filter medium body in order to respectively attach an end disk to the filter medium body.

The closed end disk **8** at the narrower side of the cone-shaped filter medium body **7** has a plurality of radially extending tooth projections **8A** such the closed end disk **8** corresponds to the outer contour of the filter medium body **7** at the narrow end face. Thus, no end disk material of the end disk **8** projects past the outer contour of the filter medium body **7** at its raw side. The compressed section **10** of the filter medium body **7** adjoins immediately the closed end disk **8**. The compressed section **10** extends axially past a length which is only slightly larger than the axial thickness of the end disk **8**.

The oppositely positioned end face of the filter medium body **7** is covered by the open end disk **9** which comprises a large central cutout through which the purified fluid can axially flow out of the interior of the filter medium body **7**. Accordingly, the open end disk **9** is of an annular shape. The outer contour of the open ring-shaped end disk **9** forms a circular ring and projects slightly past the outer contour of the filter medium body **7** at its wider end face. The inner contour of the open end disk **9** has a plurality of radially inwardly extending tooth projections **9A** such that the open end disk **9** is matched to the inner contour of the filter medium body **7** at its wider end face. In this way, it is achieved, on the one hand, that the end faces of the filter pleats of the filter medium body **7** are covered flow-tightly by the end disk at the open end face. On the other hand, a maximum free flow area at the open end face of the filter medium body **7** is provided so that the purified fluid can flow axially out of the interior of the filter medium body **7** without any obstacles.

The open end disk **9** is located at the wider end face of the filter medium body **7** and frames the filter medium body **7** in the end face section. A compressed section **12** adjoins here axially which ensures that in the manufacturing process of the round filter element **3** no liquid or flowable end disk material can spread into the compressed section **12** of the filter medium body **7**.

At the outwardly positioned side face of the open end disk **9**, a circumferentially extending sealing ring is arranged; the sealing ring position **11** is illustrated in FIG. **4**. The sealing ring at the sealing ring position **11** ensures flow-tight separation of the clean side from the raw side of the filter element **3**. In the installed state, the round filter element **3** is

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supported with the sealing ring loaded in axial direction at the sealing ring position **11** at a housing-associated component of the filter housing.

What is claimed is:

1. An annular filter element, comprising:

an annular filter medium body of a filter medium surrounding a longitudinal axis (L), the annular filter medium body comprising:

a first axial end face;

a first end disk arranged on the first axial end face;

a second axial end face arranged axially opposite to the first axial end face;

a second end disk arranged on the second axial end face; and

a plurality of circumferentially spaced folds formed into the filter medium, forming a plurality of pleats extending from the first axial end face to the second axial end face;

wherein at least one of the first and second end disks is identically embodied or at least substantially identically embodied to a radially outer or radially inner contour of the plurality of pleats, so as to form a plurality of radially extending tooth projections in the at least one of the first and second end disks, forming substantially a shape of a toothed gear;

wherein at least one of the first and second end disks is injection molded directly onto a respective one of the axial end faces of the filter medium body; and

wherein the annular filter medium body has at least one circumferentially extending axially compressed section of the filter medium arranged adjacent to and spaced axially away from a nearest one of the first and second axial end face of the annular filter medium body, forming an uncompressed free region of the filter medium between the axially compressed section and the nearest one of the first and second axial end face of the annular filter medium body.

2. The annular filter element according to claim 1, wherein the first end disk is a closed end disk, closing over a radial interior of the annular filter medium body.

3. The annular filter element according to claim 2, wherein

the first end disk is identically embodied or at least substantially identically embodied to the radially outer contour of the plurality of pleats, so as to form a plurality of radially outwardly extending tooth projections in the first end disk.

4. The annular filter element according to claim 1, wherein

the second end disk is an open end disk having a central opening that opens into a radial interior of the annular filter medium body,

wherein the second end disk is identically embodied or at least substantially identically embodied to a radially inner contour of the plurality of pleats, so as to form a plurality of radially inward projecting tooth projections in the second end disk.

5. The annular filter element according to claim 1, wherein

pleat tips of the plurality of pleats of the annular filter medium body taper axially inwardly towards the longitudinal axis (L), forming a cone-shaped annular filter medium body.

6. The annular filter element according to claim 4, wherein

the second end disk having the central opening arranged on the second axial end face of the annular filter

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medium body has a larger radial diameter than the first end disk arranged on the first axial end face.

7. The annular filter element according to claim 4, wherein

a circumferentially extending seal ring of an elastomeric material is arranged on a radially outer side of the second end disk.

8. An annular filter element, comprising:

an annular filter medium body of a filter medium surrounding a longitudinal axis, the annular filter medium body comprising:

a first axial end face;

a first end disk arranged on the first axial end face;

a second axial end face arranged axially opposite to the first axial end face;

a second end disk arranged on the second axial end face; and

a plurality of circumferentially spaced folds formed into the filter medium, forming a plurality of pleats extending from the first axial end face to the second axial end face;

wherein at least one of the first and second end disks is identically embodied or at least substantially identically embodied to a radially outer or radially inner contour of the plurality of pleats, so as to form a plurality of radially extending tooth projections in the at least one of the first and second end disks, forming substantially a shape of a toothed gear;

wherein at least one of the first and second end disks is injection molded directly onto a respective one of the axial end faces of the filter medium body;

wherein the annular filter medium body has at least one circumferentially extending axially compressed section of the filter medium arranged adjacent to and spaced axially away from a nearest one of the first and second axial end face of the annular filter medium body, forming an uncompressed free region of the filter medium between the axially compressed section and the nearest one of the first and second axial end face of the annular filter medium body; and

wherein the axially compressed section forms a barrier separating plastic material of the first or second end disk nearest the axially compressed section away from the uncompressed free region of the filter medium, the barrier forming a stop barrier of the injection molded plastic material.

9. A method for producing the annular filter element according to claim 8, the method comprising the steps of:

forming the plurality of pleats into the filter medium;

wrapping the filter medium about the longitudinal axis (L), to form the annular filter medium body;

wherein the method further comprises:

forming the axially compressed section by axially compressing the filter medium so as to form the axially compressed section spaced away from the nearest axial end face of the annular filter medium body, such that the annular filter medium body has the uncompressed free region of the filter medium arranged between the axially compressed section and the nearest axial end face of the annular filter medium body;

providing a flowable plastic material;

injection molding the flowable plastic material onto opposing axial end faces of the annular filter medium body forming the end disks;

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wherein the injection molding step further comprises the step of:

forming a plurality of radially extending tooth projections onto a radially inner side or radially outer side of the end disk, forming substantially a shape of a toothed gear;

wherein the radially extending tooth projections substantially follow a radially outer or radially inner contour of the plurality of plurality of pleats, forming a plurality of radially extending tooth projections.

10. The method for producing the annular filter element according to claim 9, wherein

the step of forming the axially compressed section further comprises the step of:

arranging the axially compressed section at distance of at least 5 mm away from the nearest axial end face of the annular filter medium body.

11. The method for producing the annular filter element according to claim 9, wherein

the step of injection molding is practiced either by:

immersing the uncompressed free region of the annular filter medium body into the flowable plastic material, or

injection molding the flowable plastic material onto the uncompressed free region of the annular filter medium body.

12. The method for producing the annular filter element according to claim 9, wherein

the step of injection molding is practiced either by:

immersing the uncompressed free region of the annular filter medium body into the flowable plastic material, or

injection molding the flowable plastic material onto the uncompressed free region of the annular filter medium body.

13. The method for producing the annular filter element according to claim 10, wherein the step of wrapping the filter medium further comprises the step of:

forming the annular pleated filter medium body to have an outer diameter that tapers axially inwardly towards the longitudinal axis (L), forming a cone-shaped annular filter medium body.

14. The method for producing the annular filter element according to claim 11, wherein

in the step of forming the annular pleated filter medium body, wrapping the filter medium such that the first axial end face has a smaller outer diameter than the opposite second axial end face such that the annular pleated filter medium body has an outer diameter that changes conically from the first axial end face to the opposite second axial end face at a cone angle of maximally 30 degrees.

15. The method for producing the annular filter element according to claim 9 wherein

the injection molding step includes:

molding the first end disk to be identically embodied or at least substantially identically embodied to the radially outer contour of the plurality of pleats, so as to form a plurality of radially outwardly extending tooth projections in the first end disk conforming to the radially outer contour of the plurality of pleats.

16. The method for producing the annular filter element according to claim 9 wherein

the injection molding step includes:

molding the second end disk as an open end disk having a central opening that opens into a radial interior of the annular filter medium body;

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the molding step includes the step of
forming the plurality of radially inward projecting
tooth projections conforming to a radially inner
contour of the plurality of plurality of pleats.

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